1.

Which of the following is a common application of tree data structures in computer operating systems?

A. Representing the connections in a computer network (e.g. , routers and switches).
B. Managing the flow of data packets in a network.
C. Organizing the file system like directories and subdirectories on a disk.
D. Storing the contents of individual files on the hard drive.

Answer: C

2.

What is a full binary tree?

A. A binary tree in which every node has either zero or two children.
B. A binary tree in which every node has either zero or one children.
C. A binary tree in which every node has either one or two children.
D. A binary tree where each node has exactly two children.

Answer: A

3.

Which of the following is a self-balancing Binary Search Tree?

A. Binary Tree
B. Binary Search Tree
C. B-tree
D. AVL Tree

Answer: D

4.

If the in-order traversal of a binary tree is D, B, E, A, G, C, F and its pre-order traversal is A, B, D, E, C, G, F.

what would be the post-order traversal of this tree?

A. D, E, B, C, F, G, A
B. D, E, B, G, F, C, A
C. D, E, A, G, B, C, F
D. G, F, C, E, D, B, A

Answer: B

5.

A graph where every vertex is connected to every other vertex by a unique edge is called ______ graph.

A. Bipartite
B. simple
C. Complete
D. Connected

Answer: C

6.

Which of the following is the example of practical applications of graph data structures?

A. Processing speed of network devices.

B. Storing user passwords securely.

C. Representing friendships and connections between users.

D. User browsing history.

Answer: C